



Technical Manual

Inert gas fire suppression system
detex Ar

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1 About the manual

This technical manual contains the information necessary for the installation and maintenance of the Protecfire fire suppression system – **detex Ar** for variety of electrical equipment such as electrical control cabinets, transformers, server cabinets and etc.

All requirements given in this manual, such as limits specified for distances, number of nozzles etc., must be adhered to in order for the fire suppression system to fulfil its purpose.

The manual explains all the components that can be used in fire suppression systems for electrical equipment of different sizes. The components installed in your fire suppression system must be recorded in this manual on the *Installed components* page. The manual is also used to document maintenance work. It contains forms prepared for this purpose.

The manual must be easily accessible but kept away from areas at risk of fire. This enables you to look up how to put the fire suppression system back into operation quickly after deployment without unnecessary loss of time. Suitable storage locations include the document cabinet in which the equipment manual is stored.

We reserve the right to make changes that serve technical progress without notice while retaining the essential features of the described system type. Likewise, Protecfire reserves the right to make changes to the contents of this manual without having to inform third parties.

2 Basic safety information

There is no pressure throughout the pipeline during standby mode when the machine is operating, this is the uniqueness of Protecfire fire suppression system. Therefore, only the usual safety rules must be observed during installation as far as the fire suppression system is concerned.

2.1 Intended use

The fire suppression system described in this technical manual has been developed for use as a fixed installed fire suppression system in the electrical equipment.

The fire suppression system is specially designed for suppressing the following fires:

- Fires involving solid materials (fire class A). These include fires involving plastics, wood products, natural rubber, wood chips, cables and etc.
- Fires involving liquid, non-polar hydrocarbons (fire class B). These include fires involving fuels, lubricating oil, hydraulic oil and etc.
- Fires involving live electrical equipment (fire class E - Australia) / fire class C - United States). These include electrical fires involving in electrical control cabinets, transformers, server cabinets and etc.

The fire suppression system may only be used as follows:

- In proper, technically perfect condition
- Without unauthorized changes
- Planned, installed and maintained by Protecfire or by personnel trained and certified by Protecfire

Modifications and changes to the fire suppression system may only be conducted by Protecfire or by personnel trained and certified by Protecfire and in consultation with Protecfire beforehand. Modifications or changes without prior consultation will result in complete exclusion of liability and loss of warranty.

The fitting or installation of any additional equipment that affects or supplements the functions of the fire suppression system requires the prior written approval of Protecfire. If necessary, a permit from the local authorities must also be obtained.

2.2 Operator obligations

For the purposes of this technical manual, the term “Operator” refers to any natural or legal person who uses the fire suppression system themselves or on whose behalf it is used. In special cases (e.g. leasing, rental), the operator is the person who must observe the aforementioned operational obligations in accordance with the existing contractual agreements between the owner and the user of the fire suppression system.

The operator must ensure that the fire suppression system is only used for its intended purpose and that hazards of any kind to the life and health of the user or third parties are avoided. The operator is responsible for ensuring that all rules listed in this manual governing the safe handling of the fire suppression system are followed.

The operator must ensure that the fire suppression system is approved and monitored in accordance with the legal regulations of the country in which the fire suppression system is operated. In addition, it must be ensured that the accident prevention regulations, the regulations of the employers' liability insurance association, the TÜV, the VDE or corresponding national provisions are observed.

Maintenance work and the replacement of components must only be conducted by Protecfire or by personnel trained and certified by Protecfire. Maintenance must be documented by the operator in the technical manual. It is the responsibility of the operator to commission Protecfire or personnel trained and certified by Protecfire with the maintenance in good time.

The operator must be familiar with the operation and handling of the fire suppression system.

Any modification in the protected area that may constitute an obstacle to the fire suppression system must be agreed in advance with the installer.

3 Extinguishing agent properties

The inert gas fire suppression systems **detex Ar** put out fires by flooding the protected area with argon to reduce oxygen levels.

Argon is a noble gas derived from the ambient air and by volume of up to 0.93 vol% part of the natural atmosphere. Its density in relation to air is 1.38 to 1. Argon as natural components of the ambient air are non-toxic and electrically non-conductive. As a natural component of ambient air, argon is easily available, non-toxic, non-conductive, and environmentally friendly, making it a sustainable resource.

Due to its heavier-than-air properties, argon is particularly suitable for top-down release, making it effective for Class A (solid material) and Class B (flammable liquid) fires.

Additionally, argon does not conduct electricity, making it suitable for areas with electrical or electronic components, such as electrical control cabinets, cable bridges, and small spaces with high-value equipment. Argon, like other fire extinguishing agents such as gas, water, foam, or dry powder, can be used to extinguish fires without leaving any residue. After extinguishing a fire with argon, the area only needs to be ventilated to allow the extinguishing agent to evaporate, leaving no residue behind.

The **detex Ar** inert gas fire extinguishing system is non-toxic even at a gas concentration that is effective for extinguishing fires. However, it reduces oxygen concentrations compared to ambient air. Therefore, all individuals should be reminded to stay away from the fire extinguishing area before and after the fire extinguishing process.

Features & Advantages:

- No residue after extinguishing the fire
- No damage to the equipment being protected
- Suitable for high and low ambient temperature environments
- Fast filling, convenient and economical
- Non-conductive, suitable for high-precision electronic equipment
- Inert gas, chemically neutral
- Non-oxidizing
- Poses no threat to the environment (Zero ozone depletion / zero global warming)

4 Functions and mode of operation

4.1 What are the features of fire extinguishing systems?

This innovative gas fire suppression system was developed especially for use under harsh climatic conditions (operating temperatures from - 50 °C to + 65 °C). In particular, for the protection of enclosed equipment, such as switch and distribution cabinets, cable trays, machine tools (equipment protection) or closed areas with generators or internal combustion engines.

The fire suppression solution has been developed as a pre-dimensioned kit, so that hydraulic calculations for the line lengths and pipe diameters are no longer required. Here, the limits specified by the manufacturer for distances, pipe diameter, line lengths, pressures, operating temperatures etc. must be strictly adhered to.

Thanks to their intelligent design, the inert gas fire suppression systems **detex Ar** is particularly compact with space-saving storage of the extinguishing agent.

The extinguishing agent, argon is stored compressed in high pressure steel cylinders at 150 bar at 20 °C. Three standard cylinder sizes (10 L, 20 L, 50 L) are available to make installation options are highly versatile and always feasible, even in very tight spaces. The 150 bar technology allows the use or installation of these suppression systems even in extreme temperature conditions, from - 50 °C to + 65 °C.

The service life of extinguishing agent containers (excluding fire extinguishing agents) is up to 10 years and easy to maintain. Once activated, they will need to be promptly returned to the factory for servicing or replacement to ensure that the fire extinguishing system can be operated again.

WARNING:

The gas container must be refilled by Protecfire or by dealers approved by Protecfire

Due to the gas nozzles which were specifically developed by Protecfire for this suppression system, it is possible to reduce the discharge pressure well below 25 bar during activation, so that a nearly constant extinguishing agent flow can be achieved without pressure peaks. This reduces unwanted vibrations on the areas to be protected.

Other features:

- **No external power supply is required:** Protecfire fire suppression systems operate pneumatically - mechanically. This means that they do not require any additional power supply.
- **No pressure in the pipeline:** The fire suppression system is unpressurized in the pipeline until it is activated.
- **Ease of Maintenance:** Major components are needed to replace up to 10 years period.
- **Automatic and / or manual activation / or remote activation:** Automatic activated by SPY fire detection elements. The fire suppression system can also be activated by pneumatic manual actuator, or remotely activated by E-SPY electric manual actuator (optional).

4.2 How does fire detection work?

Protecfire uses thermos-pneumatic SPY fire detection elements to detect fire. These elements only activate the fire suppression system if a specified temperature is exceeded. Failure due to the failure of electrical components is thus ruled out. Fire detection elements with different activation temperatures can be used on equipment with thermally inhomogeneous area or it contains different temperature zones.

4.3 What status indications are present on the fire suppression system?

Protecfire **detex Ar** fire suppression system is pressurized in the extinguishing container when it is in standby mode. For this reason, it is necessary to monitor the container pressure by observing the pressure gauge.

4.4 What happens in the event of a fire?

detex Ar fire suppression system consists of high-pressure gas cylinder which filled with argon gas at 150 bar, pneumatic manual actuator, SPY fire detection element, nozzle, control line and extinguishing line and other optional parts. When a fire detection element is heated to its activation temperature, a mechanism opens the built-in argon propellant gas cartridge. This causes an increase in pressure in the control line. The pressure control valve in the high-pressure gas cylinder will open to release compressed argon gas through the line to the nozzles, completes fire suppression process. If the fire temperature does not reach its activation temperature, the fire suppression system can be activated by operating the pneumatic manual actuator. If the E-SPY electric manual actuator has been installed,

the fire suppression system can be activated by remote electrical signal.

The extinguishing effect of the inert gas argon (Ar) is mainly due to the reduction of the oxygen content of the air so that the combustion process cannot continue. The gas passes through the flood area quickly, providing complete compartment protection. It is much heavier than air and is therefore ideal to install at high positions of the compartment.

An advantage of providing the extinguishing agent in a gaseous form is that there is no evaporation as it flows out. This means that there is neither mist nor condensation, which, in combination with breakdown products from the fire, might cause consequential damage (especially in electrical rooms). The "cold shock", which is a constant risk with other gas systems is eliminated.

4.5 Can the system shut down the equipment in the event of a fire?

Yes. **detex Ar** fire suppression system includes a linkage function (alarm and shutdown). When there is fire, the fire suppression system is activated, pressure switch works at the same time to generate signal to the alarm to indicate the fire suppression system is being activated.

The following components must be installed for these additional optional operations:

- Alarm Controller
- Pressure Switch

In addition, the devices that are to be switched off or on must be properly connected to the alarm controller.

The system operates independently to other electrically operated components. Power for the pressure switch is supplied via the connection to the electrical system on the equipment.

NOTE:

Protecfire supplies the alarm controller and pressure switch and does not install wiring.

Customer shall commence wiring works by qualified electrical personnel.

4.6 General technical data

Fire classes	A (fires involving solid materials including plastics, wood products, natural rubber, wood chips, cables and etc.) B (fires involving liquid, non-polar hydrocarbons such as fuels, lubricating oils, hydraulic oils and etc.) Electrical (Fires involving live electrical equipment)
Extinguishing agent container	High pressure steel extinguishing agent container in standard sizes 10 L, 20 L, 50 L.
Mode of activation in the event of a fire	Automatically activated by SPY fire detection elements. Manually activated by pneumatic manual actuator. Remotely activated by E-SPY electric manual actuator (optional).
Activation speed	Ultra-fast response RTI 20
Pressure in standby mode	No pressure in Detection and extinguishing line Gas cylinder 150 bar at 20 °C
Extinguishing agent	Argon Gas
Extinguishing agent supply	Through nozzles. The number of nozzles installed depends on the volume of the object to be protected.
Signal function	Provides a signal through the pressure switch (optional) to indicate whether the fire suppression system is in standby mode or has been activated.
Maintenance	The information can be found in chapter 7 - <i>Maintenance</i> .

5 System components

The fire suppression system is designed to protect objects of different sizes and types. The extinguishing agent containers are therefore available in different sizes. In addition, the fire suppression system is adapted to customer requirements. Therefore, not every component presented in this chapter needs to be installed in your system. The components that are installed in your fire suppression system can be found in the section *Installed components*.

Product Selection Guide



With Argon

★Standard product

Part Number

Protecfire (Langfang)

Product Category Code

detex Ar

Product Generation

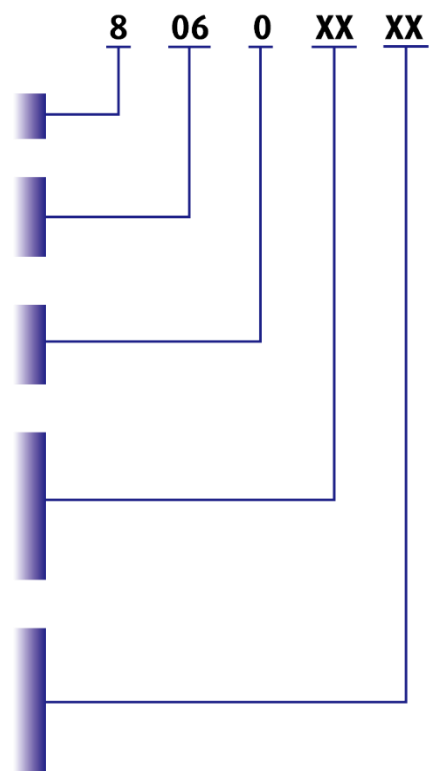
detex Ar (Model 1.00)

Container Size

10 = Container 10 L
20 = Container 20 L
50 = Container 50 L

Operation Pressure

00 = 0 Bar
15 = 150 bar (★Standard product)
D = DEMO Samples



NOTES:

- DEMO Samples: without extinguishing agent, without argon, with thread glue, with sample sticker, with sample laser marking.

5.1 Extinguishing agent container

The extinguishing agent container contains a permanently installed pressure gauge, the extinguishing agent, pressure switch (optional) and a pressure control valve. The control valve consists of an activation mechanism and pressure control device. When a fire detection element or a manual actuator / or E-SPY electric manual actuator (optional) is activated, pressure rises rapidly in the pipeline and reaches to the extinguishing agent container via the lines. The increased pressure will activate the mechanical mechanism in the pressure control valve on the extinguishing agent container to release the extinguishing agent through the extinguishing line to the nozzles.

Extinguishing agent containers are available in different sizes (see table below). The argon gas cylinders are uniquely color coded to allow for quick and easy identification.



Argon gas extinguishing agent container specification

Cylinder capacity [Liter]	Diameter [mm]	Height [mm]	Weight with Filling [kg]	Filling pressure [bar]	Mass of agent [kg]	Volume of agent 20 °C 101 Kpa [m ³]	Ref. max. protected space 38% 21 °C [m ³]
10	140	1059	19.5	150	2.71	1.52	2
20	203	1034	34.7	150	5.42	3.04	4
50	232	1669	68.5	150	13.56	7.60	10

NOTE:

- Never assume an extinguishing agent container is empty. Treat all the extinguishing agent containers as if they were fully charged.
- Local laws and regulations must be observed, especially regarding pressure equipment directives.
- Precise adjustment of the control valve must be considered for proper function. And tighten the necessary safety accessories.

5.1.1 Installation

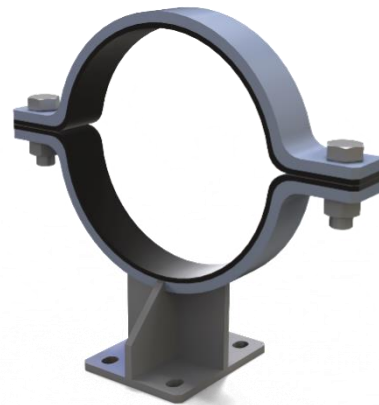
General planning and installation rules

- Extinguishing agent containers must be installed outside the area designated for fire extinguishing and fixed directly by using two band clamps.
- The installation location must be selected so that the extinguishing agent container is not exposed to high temperature and any additional forces and / or stresses.
- If Protecfire container band clamps are not used, the following requirements must be observed:
 1. The band clamps must be designed for five times the weight of the container.
 2. No steel on steel: Intermediate rubber or plastic inserts must be placed between the clamps and the extinguishing agent container.

Band clamp information



Band clamp (Light duty)
for 10 L, 20 L, 50 L container



Band clamp (Heavy duty)
for 50 L container

Name	Part number	Installation tool
Band clamp (Light duty) for container 10 L	85314010	Wrench size 13 mm, 17 N·m
Band clamp (Light duty) for container 20 L	85312214	Wrench size 13 mm, 17 N·m
Band clamp (Light duty) for container 50 L	85314050	Wrench size 13 mm, 17 N·m
Band clamp (Heavy duty) for container 50 L	85314150	Wrench size 18 mm, 50 N·m

5.1.2 Control valve

Technical features

Extinguishing agent container thread	25E - EN ISO11363-1
Filling connection	W21.8 x 1/14"
Pipe connection	M18 x 1.5 / M14 x 1.5
Safety bursting disc device	270 bar
Temperature range	- 50 °C to + 65 °C
Operating pressure	max. 200 bar
Extinguishing agent	Argon
Material of valve body	Brass / Stainless steel



5.2 SPY fire detection element

The fire suppression system can be automatically activated using the SPY fire detection element.

The SPY fire detector element is a purely mechanical device that automatically detects temperature and will not fail to activate the fire suppression system due to faulty electrical components. When a fire is detected, the fire detection element heats up to the activation temperature as the temperature within the protected area increases. The heat-sensitive pressure-releasing element bursts, the built-in argon cylinder is opened, and the pressure in the pipeline rises rapidly. This activates the extinguishing agent container and starts the rapid suppression process. The RTI value (Response Time Index) is 20, providing a very high-speed reaction rate.

The activation temperature is determined in consultation with the manufacturer or customer, according to the conditions of the fire hazard area. The activation temperature is based on the working temperature of the protection target, but is set to be at least 30 °C higher than the maximum working temperature of the protection target.



SPY fire detection element

Technical data

Applicable pipeline length	≤ 20 m (DN 6)
Response rate	Ultra - fast response RTI 20
Material	Stainless steel
Weight	366 g
Connection	M14 x 1.5 / Torque 30 N·m
Maintenance	Regular visual inspection. Replace it every 5 years
After activated	Replacement

Activation temperature & color code

Part number	Activation temperature	Color code
82105700	57 °C (134 °F)	Orange
82106800	68 °C (154 °F)	Red
82107900	79 °C (174 °F)	Yellow
82109300	93 °C (199 °F)	Green
82111000	110 °C (230 °F)	Blue
82114100	141 °C (286 °F)	Navy Blue
82118200	182 °C (359 °F)	Purple
82123000	230 °C (446 °F)	Grey
82126000	260 °C (500 °F)	Black

5.2.1 Installation

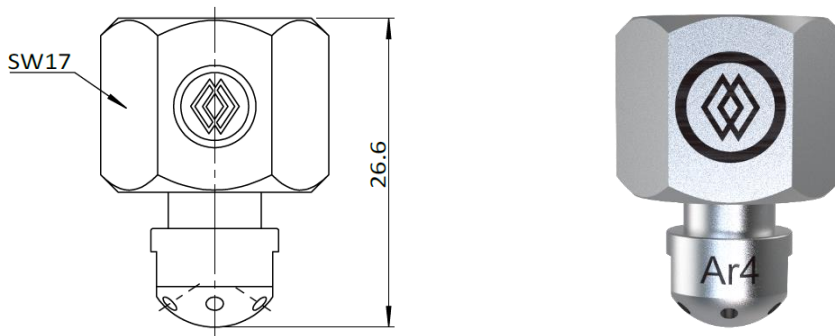
- The fire detection element can be installed in any position in relation to the object to be protected. However, the thermal effect must not be hindered by fixtures or equipment. Air currents that can hinder the thermal effect must be taken into account during positioning.
- The number of fire detection element is determined by the Protecfire manufacturer / customer according to local conditions and the number of targets need to be protected.
- Each fire detection element monitors a maximum area of 1.0 m².
- Distance from object to be protected: minimum 300 mm to maximum 2000 mm.
- Use wrench size 17 mm and apply torque 30 N·m to connect to pipeline.

WARNING:

Only skilled and trained & certified person may install, uninstall or maintain. In order to prevent the system from being started accidentally, the system must be shut down and the control line removed during dis-assembly and maintenance.

5.3 Ar4 gas nozzle

The Ar4 gas nozzle is specially designed for excellent gas discharge efficiency with particularly small nozzle holes to achieve extremely fine and consistent release of fire extinguishing agent. Ar4 nozzle is one of the smallest and most robust of its kind. Due to the unique design structure, the gas nozzle is insensitive to coarse dirt and it will not affect the gas release from the nozzle.



Protective caps

In order to ensure proper activation of the fire extinguishing system, the Ar4 nozzle must be equipped with a red protective cap in perfect condition. The red protective cap can also protect the Ar4 nozzle from contamination.

The protective cap is made of permanently elastic silicone rubber with high temperature resistance up to 200 °C. Its material will not bond to the stainless steel nozzle. After the extinguishing agent container is activated, the extinguishing agent travels through the extinguishing line to the nozzle and the protective cap will be ejected when the activated pressure reaches to 2 bar or above.

Technical data

Operating temperature	-50 °C to +200 °C
Nozzle material	Stainless steel
Protective cap material	Silicone rubber
Medium	Argon
Weight	28 g
Connection	M14 x 1.5 / Torque 30 N·m
Part number	85014004

5.3.1 Installation

General planning and installation rules

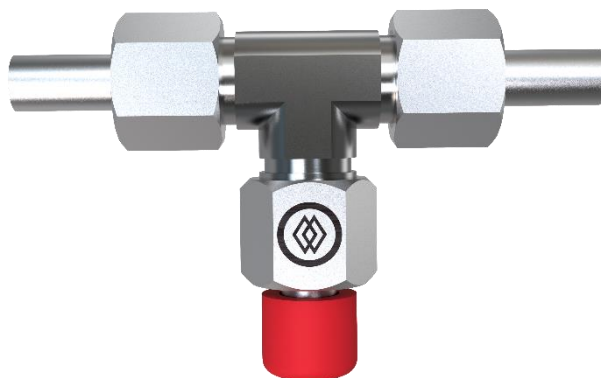
- The nozzles must be installed in such a way that all protected areas are evenly covered with extinguishing agent.
- As argon is significantly heavier than air and is thus particularly suitable to be installed at a higher position inside the protected area.
- Connection: M14 x 1.5, 30 N·m.

NOTE:

To determine the number of nozzles in the system is depended on the gross volume of the protected area.

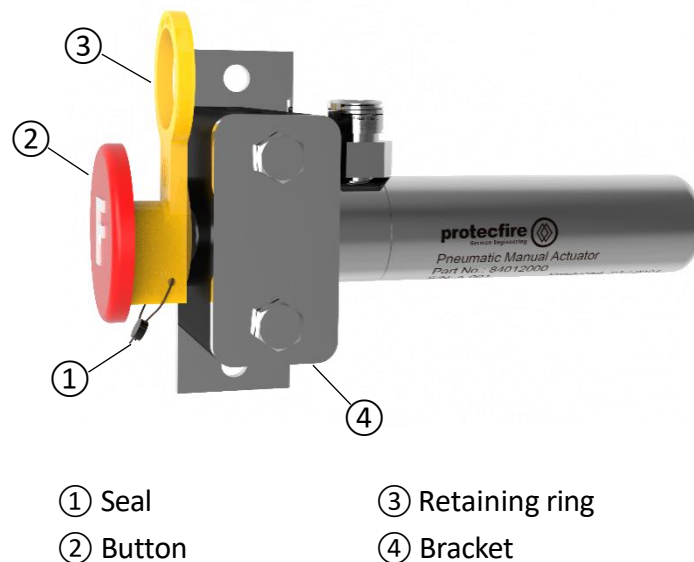
Mounting the nozzles

The focus of the development was spraying the extinguishing agent perfectly and covering the target evenly. Ease of assembly was also an important criterion in the selection and development of its form. Protecfire opts to use nozzle that can be installed easily without additional components. This makes assembly much simpler.



5.4 Pneumatic manual actuator

The fire suppression system can be manually activated by using the pneumatic manual actuator. When the red button is struck, the built-in argon cylinder is opened, and the pressure in the pipeline rises rapidly. This activates the extinguishing agent container and starts the rapid suppression process. The pneumatic manual actuator must be replaced after use or if the seal is damaged.



Procedure in the event of a fire

1. Pull the retaining ring ③ and destroy the seal ① at the same time.
2. Strike the red button ②. The fire suppressing process starts.

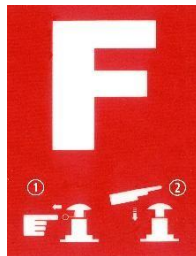
Technical data

Applicable pipeline length	≤ 20 m (DN 6)
Material	Stainless steel
Weight	585 g
Release force	70 N
Connection	M14 x 1.5 / Torque 30 N·m
Maintenance	Regular visual inspection. Replace it every 5 years
After activated	Replacement
Included items	Manual actuator device, bracket, information sign
Part number	84012000

5.4.1 Installation

General planning and installation rules

- The number of pneumatic manual actuator is determined by the Protecfire manufacturer / customer according to local conditions and the number of targets need to be protected.
- The pneumatic manual actuator can be installed in any orientation.
- The installation location must not be in close proximity to a potential fire source and must be easily accessible in the event of a fire.
- The information sign supplied must be attached near the manual actuator to indicate clearly the location of the pneumatic manual actuator and to provide clear instructions to the operator.



Information sign of the pneumatic manual actuator

- Use wrench size 17 mm and apply torque 30 N·m to connect it to pipeline. When installing, secure with a second wrench to prevent rotation.

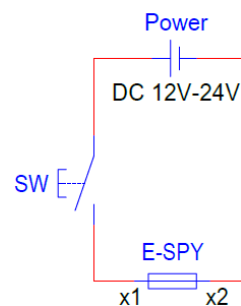
5.5 E-SPY electric manual actuator (optional)

The E-SPY electric manual actuator is a remote activation device used in Protecfire automatic fire suppression system, which is activated by remote electrical signal.

The electric manual actuator is connected to the DC 12 V - 24 V power supply, when the current flows through the heat-sensitive pressure-releasing element, it starts to heat up. When the temperature is reaching its activation temperature, the heat-sensitive pressure-releasing bursts and the built-in argon cylinder is opened to release argon to the pipeline, the extinguishing agent container is activated for suppression process.

The E-SPY electric manual actuator must be replaced when any of the following occurs:

- After it has been activated
- Indicator pin pops out and green indicator label sticker is punctured



E-SPY electric manual actuator

WARNING:

Only skilled and trained & certified person may install, uninstall or maintain. In order to prevent the system from being started accidentally, the system must be shut down and the control cable must be disconnected during dis-assembly and maintenance, and make sure that no current is running through the electrical connections.

Technical data

Rated voltage	DC 12 V - 24 V
Max. current	1.5 A
Operating temperature	- 55 °C to activation Temp. + 79 °C
Applicable pipeline length	≤ 20 m (DN 6)
Material	Stainless steel
Cable	2 x 0.75 mm ² x 2.5 m, diameter approx. 6.0 mm
Weight	Approx. 1.1 Kg
Connection	M14 x 1.5 / Torque 30 N·m
Maintenance	Regular visual inspection. Replace it every 5 years
After activated	Replacement
Included items	E-SPY electric manual actuator, bracket
Part number	84107900

5.5.1 Installation

General planning and installation rules

- During installation, the green indicator label sticker of the E-SPY electric manual actuator should be clearly visible for the purpose of routine inspection.
- The installation location of the E-SPY electric manual actuator must not be in close proximity to a potential fire source, and the ambient temperature must not exceed 79 °C to avoid accidental activation.
- The E-SPY electric manual actuator can be installed on the same pipeline with the pneumatic manual actuator that connects to the extinguishing agent container.
- The number of E-SPY electric manual actuator is determined by the Protecfire manufacturer / customer according to local conditions and the number of targets need to be protected.
- Use wrench size 17 mm and apply torque 30 N·m to connect to pipeline.

NOTE:

Protecfire supplies the E-SPY electric manual actuator and does not install wiring.

Customer shall commence wiring works by qualified electrical personnel.

5.6 Extinguishing line and control line

detex Ar fire suppression systems is set up as a single-line system only, where the automatic detection and activation function is combined with the fire extinguishing function to create a combined automatic activation and extinguishing line. This design feature is unique to Protecfire fire suppression systems and is called the detexline.

The name detexline consists of abbreviations for these functions:

- det (detection) stands for fire detection function through connected fire detection elements.
- ex (extinguishing) stands for the fire extinguishing function through connected extinguishing nozzles.
- line (pipeline) stands for the combination of these functions in a line.

The extinguishing line connects fire detection elements and extinguishing nozzles to the extinguishing agent container. The extinguishing line consists of a hose and a pipe. The hose connects between the extinguishing agent container and the pipe. Its flexibility protects against vibration and thermal expansion.

As a complete automatic fire extinguishing system, in addition to the detexline line, the detex Ar fire extinguishing system has an additional manual control line, referred to as the control line. The control line connects between the pneumatic manual actuator and / or E-SPY electric manual actuator and the detexline pipeline. The control line consists of a hose / or a pipe.

The Extinguishing line & Control line is recommended to be secured by using R-type pipe clamps.

The length of control line and extinguishing line depends on the system structure, the maximum length specified in the following installation instructions must not be exceeded.

Technical data

Dimensions of extinguishing line	Pipe: Outer Diameter 8 mm; Pipe thickness 1 mm Hose section: DN 6
Dimensions of control line hose	Pipe: Outer Diameter 8 mm; Pipe thickness 1 mm Hose section: DN 6
Materials of pipe section	Stainless steel
Materials of hose section	Teflon stainless steel braided

Connection to extinguishing line	M14 x 1.5 / Torque 30 N·m
Connection to control line	M14 x 1.5 / Torque 30 N·m
Pressure	No pressure in standby mode, low pressure after activated
Maintenance	Regular visual inspection
Part number	According to demands

5.6.1 Installation

General planning and installation rules

- The connecting hose to extinguishing agent container must only be routed outside fire hazard areas.
- Piping should only be laid out where it will not be damaged by external influences and cannot be easily damaged.
- The pipes must be secured to the equipment by using pipe clamps.
- The pipe clamps must not be used as holders for other objects.
- The maximum distance between the pipe clamps is 600 mm.



As shown above: Maximum distance between the pipe clamps

- Hose / extinguishing agent container connection Wrench size 17 mm, 30 N·m
- Hose / pipe connection Wrench size 17 mm, 30 N·m

NOTE:

- **The maximum length of the connecting extinguishing pipeline and control line to extinguishing agent container is 20 m (DN 6). In case of special circumstances, the control line can be extended by adding a booster for longer distance. Please consult Protecfire for enquiry.**
-

5.7 Booster (optional)

The booster is a pneumatic pressurizing device used in Protecfire automatic fire suppression system.

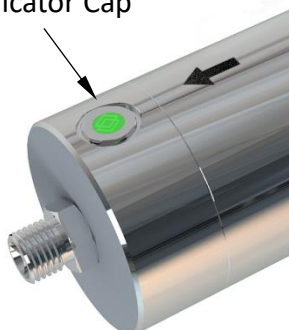
The booster releases the internal compressed gas which then increases the pressure inside the control line, after it is being activated by the pressure from the SPY fire detection element or pneumatic manual actuator or E-SPY electric manual actuator or another container of the same system. This function extends the length of the control line of the fire suppression system. Another function of the booster is to activate multiple extinguishing agent containers at the same time when there is more than one extinguishing agent container in the same system.



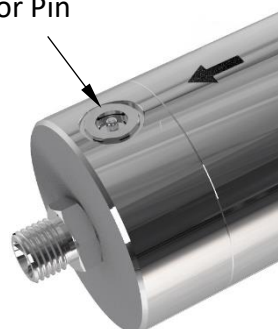
The booster must be replaced when any of the following occurs:

- After it has been activated
- The green indicator cap is displaced, and an indicator pin is visible.

Green Indicator Cap



Indicator Pin



Technical data

Specification	Booster 380
Operating temperature	- 55 °C to + 80 °C
Operating pressure	Low pressure (Argon)
Applicable pipeline length (Piping on the outlet side of the booster)	≤ 100 m (DN 6)
Material	Stainless steel
Weight	Approx. 4.5 kg
Connection	M14 x 1.5 / Torque 30 N·m
Maintenance	Regular visual inspection; Replace it every 5 years
After activated	Replacement
Included items	Booster 380, bracket
Part number	84210000

5.7.1 Installation

General planning and installation rules

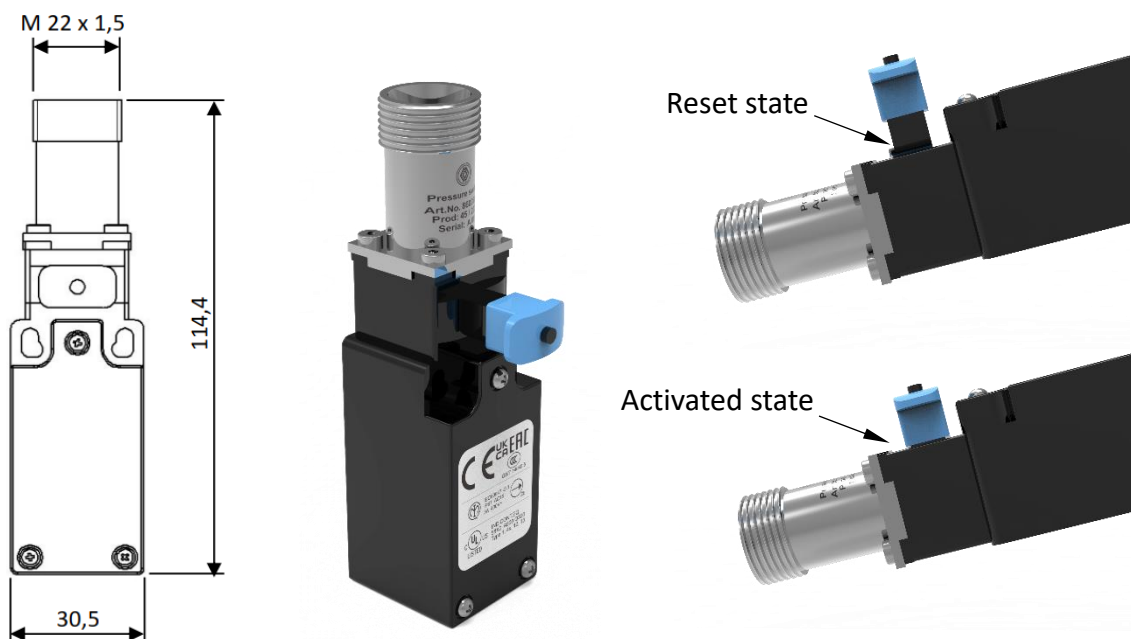
- During installation, the green indicator cap of the booster should be clearly visible for the purpose of routine inspection.
- The installation location of the booster must not be in close proximity to a potential fire source to avoid accidents.
- The number of boosters is determined by the Protecfire manufacturer / customer according to local conditions and the number of targets need to be protected.
- Use wrench size 17 mm and apply torque 30 N·m to connect to pipeline.

WARNING:

Only skilled and trained & certified person may install, uninstall or maintain. In order to prevent the system from being started accidentally, the system must be shut down and the control line removed during dis-assembly and maintenance.

5.8 Pressure switch (optional)

At customer request, **detex Ar** fire suppression system can be equipped with an optional pressure switch. Two floating contacts are provided on the pressure switch has two passive contacts (one normally closed and one normally open). It allows the objects to be protected to be disconnected from electricity and / or for an alarm signal to be generated. When the system is activated, the pressure in the pipeline increases, thus activating the pressure switch, and its blue button will move downwards and generate outputs of the signal.



If the pressure switch is properly connected to fire suppression system, the following actions are possible:

- Visual and acoustic alarm activated by the alarm controller.
- Alarm activated by external alarm devices such as horns or warning lights.
- Shut down the equipment.



Alarm Controller (optional)

NOTE:

Protecfire supplies the alarm controller and pressure switch and does not install wiring.

Customer shall commence wiring works by qualified electrical personnel.

Technical data

Switching voltage (max.)	400 V AC
Switching current (max.)	3 A AC
Protection class	IP 67
Working pressure	4.0 ± 0.3 bar
Material	Stainless steel housing, epoxy powder coated
Weight	Approx. 215 g
Connection	M22 x 1.5 / Torque 80 N·m
Part number	86006026

5.8.1 Installation

General planning and installation rules

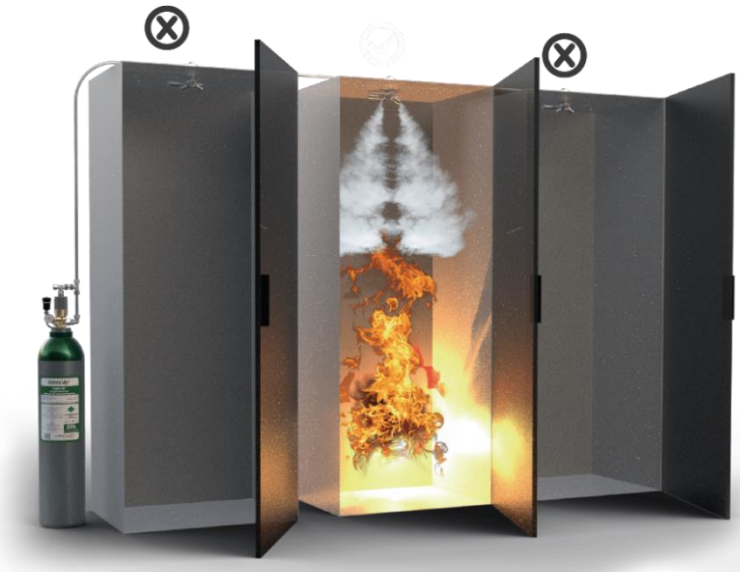
- The pressure switch is fitted onto the extinguishing pipe using an M22 x 1.5 screw joint with a tightening torque of 80 N·m.
- After activation, the blue button of the pressure switch can be lifted upward to reset it. After conducting a pressure test on the pipeline, you must check and ensure that the pressure switch has been reset. If necessary, replace the pressure switch with a new one.

WARNING:

When installing/removing the pressure switch on the container, it must be ensured that no current is running through the electrical connections.

5.9 Directional control valve (optional)

The directional control valve is a device that controls the flow direction of the medium in the pipeline. It is used in fire extinguishing systems that protect multiple zones simultaneously. By installing a directional valve, it becomes possible to accurately direct the fire extinguishing agent to the activated device (such as a fire detection element or manual actuator) that has been activated within a specific partition.



The advantages by using the directional control valve:

- Multiple compartment protection
- Simple and reliable
- Using smaller extinguishing agent container makes possible
- Accurately direct the fire extinguishing agent only to the fire location that the activation device has been activated.



Technical data

Operating pressure	0.3 bar to 2 bar
Operation temperature	- 50 °C to + 120 °C
Material	Stainless steel
Weight	285 g
Connection	M18 x 1.5 / Torque 60 N·m
Part number	80108050

NOTE:

- When the system is configured with directional control valve, the number of pneumatic manual actuator or E-SPY electric manual actuator is determined by the Protecfire manufacturer / customer according to local conditions.
- If Pneumatic Manual Actuator or E-SPY Electric Manual Actuator has been configured, the control line connected to it must be connected to the fire suppression line in the specific area to ensure that it protects the specific area, consult Protecfire for details.

5.10 Pressure gauge

In the system standby state, there is no pressure in the system pipeline, and the pressure in the extinguishing agent container is at 150 bar at 20 °C.

- A pressure gauge is incorporated on the extinguishing agent container for indicating and monitoring the pressure of compressed argon gas.
- General industrial applications.
- High reliability and long service life.
- Pressure indication angle exceeds 270 °.
- Protection class IP 65.
- The oil level in the pressure gauge is at 3/4 of the dial's diameter.



Technical data

Nominal diameter	40 mm
Accuracy	Level 2.5
Measure pressure range	0 bar to 400 bar
Operation temperature	- 50 °C to + 70 °C
Material	Stainless steel
Connection	M10 x 1
Part number	80665001

6 Installation and commissioning

6.1 Installed components

Order no: _____

Date: _____

Customer details

Company name: _____

Address: _____

Name of equipment: _____

Model of equipment: _____

Serial number of equipment: _____

Installation location: _____

Installation of fire suppression system

Installation performed by: _____

System serial number: _____

Installed components	Part no.	Quantity	Serial number
Container 10 L detex Ar	80601000		
Container 20 L detex Ar	80602000		
Container 50 L detex Ar	80605000		
Ar4 gas nozzle with protective cap	85014004		
Pneumatic manual actuator	84012000		
E-SPY electric manual actuator	84107900		
Booster 380	84210000		
Pressure switch	86006026		
Directional control valve	80108050		
SPY fire detection element _____ °C			
Band clamps			

6.2 Pressure test

detex Ar fire suppression system must be checked for any leaks. The test is primarily a pressure test to ensure the good seal performance in the pipelines. We suggest you conduct the pressure test once a year as part of annual maintenance. Protecfire and personnel trained and certified by Protecfire use a pressure test kit to test the extinguishing line and control line. The pressure test kit (part number 801MNT01) is needed to carry out the pressure test properly.

The pressure test kit mainly includes a pressure test device, a portable air pump, adapters, an extension hose, etc. (as shown in the figure below). The specific steps are as follows:

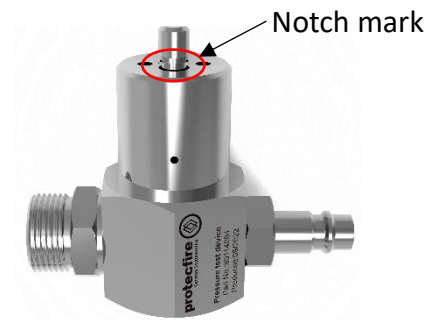


Procedure

1. Disconnect hose connection from the extinguishing agent container.
2. Cover thread on extinguishing agent container with dust proof cap to prevent contamination.
3. Select an appropriate adapter to connect the test device to the hose.
4. Connect extension hose between air pump and pressure test device.



5. Set the air pump pressure to 3 bar. Pressurize the pipeline and maintain the pressure for approximately 15 seconds. Observe the test device. If notch mark is visible, it means the test has been passed and continue the step 7.



6. If the notch mark is not visible. All the pipeline must be checked for loose connections or damage, or if the protective cap on the nozzle has fallen off. Then repeat step 5 and step 6 until the notch mark is visible.
7. Remove the test device from the hose.
8. Reconnect the hose connection back to the extinguishing agent container.
9. The test is completed.

NOTE:

When the system is configured with directional control valve, the connection between the directional control valve and the pipeline inside the compartment must be disconnected.

The pipeline inside the compartment must be tested according to the above steps.

6.3 Commissioning checklist (Initial installation)

Once all components have been assembled and the pressure test has passed, the fire suppression system can be operated normally. The checklist below is used to make a final check that the installation is correct. The fire suppression system can be handed over if all questions are answered with "Yes". Otherwise, the defect must be remedied before handover.

Extinguishing agent container

Outwardly undamaged?	Yes ☐	No ☐
Screw connections firmly tightened?	Yes ☐	No ☐
Extinguishing agent container firmly attached?	Yes ☐	No ☐
Band clamp firmly attached?	Yes ☐	No ☐

Pressure gauge

Outwardly undamaged?	Yes ☐	No ☐
The oil level in the pressure gauge is at 3/4 of the dial's diameter?	Yes ☐	No ☐
Screw connection firmly tightened?	Yes ☐	No ☐
Pressure level correct?	Yes ☐	No ☐

Pressure switch

Outwardly undamaged?	Yes ☐	No ☐
Screw connection firmly tightened?	Yes ☐	No ☐
The pressure switch in reset state?	Yes ☐	No ☐

Fire detection elements

Outwardly undamaged?	Yes ☐	No ☐
Installation in protected area?	Yes ☐	No ☐
Activation temperatures according to design?	Yes ☐	No ☐
Free of contamination?	Yes ☐	No ☐
Screw connections firmly tightened?	Yes ☐	No ☐
Pipe clamp firmly attached?	Yes ☐	No ☐

Nozzles

Outwardly undamaged?	Yes ☐	No ☐
Is there a red protective cap and is it in good condition?	Yes ☐	No ☐
Nozzles on protected object properly installed?	Yes ☐	No ☐
Screw connections firmly tightened?	Yes ☐	No ☐

Pneumatic manual actuator

Outwardly undamaged?	Yes ☐	No ☐
Easily accessible?	Yes ☐	No ☐
Mounted outside of fire hazard areas?	Yes ☐	No ☐
Retaining rings / seals undamaged?	Yes ☐	No ☐
Retaining rings free of contamination?	Yes ☐	No ☐
Information signs present and legible?	Yes ☐	No ☐
Screw connections firmly tightened?	Yes ☐	No ☐
Pipe clamp firmly attached?	Yes ☐	No ☐

Electric manual actuator

Outwardly undamaged?	Yes ☐	No ☐
Installation in specified area?	Yes ☐	No ☐
Green indicator mark intact and clearly visible?	Yes ☐	No ☐
Free of contamination?	Yes ☐	No ☐
Screw connections firmly tightened?	Yes ☐	No ☐
Pipe clamp firmly attached?	Yes ☐	No ☐

Booster

Outwardly undamaged?	Yes ☐	No ☐
Installation in specified area?	Yes ☐	No ☐
Green indicator mark intact and clearly visible?	Yes ☐	No ☐
Free of contamination?	Yes ☐	No ☐
Screw connections firmly tightened?	Yes ☐	No ☐
Pipe clamp firmly attached?	Yes ☐	No ☐

Directional control valve

Outwardly undamaged?	Yes ☐	No ☐
Screw connection firmly tightened?	Yes ☐	No ☐
Green indicator mark intact?	Yes ☐	No ☐

Extinguishing line

Outwardly undamaged and not deformed?	Yes ☐	No ☐
Pressure test passed?	Yes ☐	No ☐
Distance between pipe clamps a maximum of 0.6 m?	Yes ☐	No ☐
Screw connections firmly tightened?	Yes ☐	No ☐
Pipe clamp firmly attached?	Yes ☐	No ☐

Control line

Outwardly undamaged and not deformed?	Yes ☐	No ☐
Pressure test passed?	Yes ☐	No ☐
Distance between pipe clamps a maximum of 0.6 m?	Yes ☐	No ☐
Screw connections firmly tightened?	Yes ☐	No ☐
Pipe clamp firmly attached?	Yes ☐	No ☐

Introduction

Brief introduction to functions, relevant components and behavior in the event of fire?	Yes ☐	No ☐
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Summary

Fire suppression system handed over and ready for operation after introduction?	Yes ☐	No ☐
---	---------------------------	--------------------------

Comment

Customer's name:

Customer's signature:

Date:

Installer's signature:

Date:

7 Maintenance

7.1 Maintenance, intervals

The list below contains an overview of maintenance tasks and instructions on performing these tasks. The pages following this overview are used to document annual maintenance. Daily, weekly, monthly maintenance and half year maintenance can be carried out by the operators if they have received appropriate training beforehand. Annual maintenance must only be conducted out by Protecfire or by personnel trained and certified by Protecfire.

Daily maintenance

Implementation: Operator personnel trained in the use of the fire suppression system.

- Visual inspection of the fire suppression system.

Weekly maintenance

Implementation: Operator personnel trained in the use of the fire suppression system.

- Visual inspection of the fire suppression system.
- Check the components for damage. Pneumatic manual actuator with missing retaining rings must be replaced.

Monthly maintenance

Implementation: Operator personnel trained in the use of the fire suppression system.

- Check the nozzles, the red protective cap must be firmly seated on each nozzle and in good condition.
- Check fire detection elements, any damage on the heat-sensitive pressure-releasing element on the fire detection elements and check the presence of filling liquid inside the heat-sensitive pressure-releasing element.

Half year maintenance

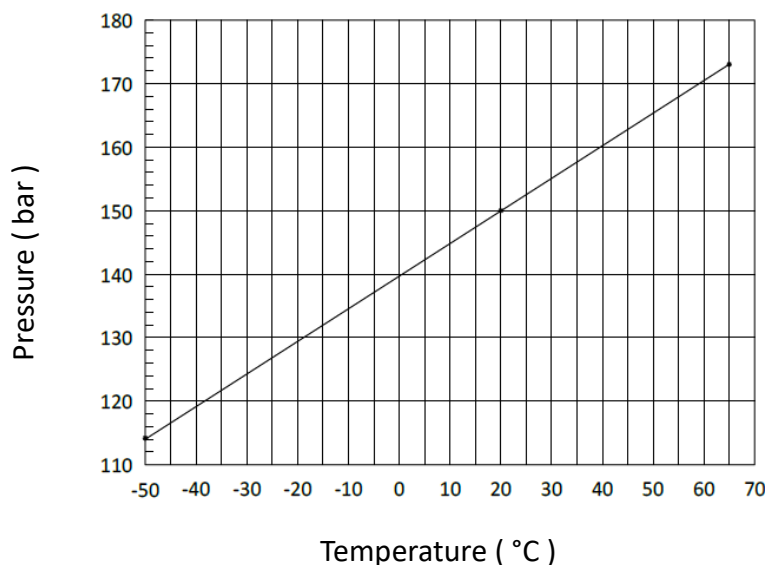
Implementation: Operator personnel trained in the use of the fire suppression system.

- Check the pressure gauge on extinguishing container. The pressure gauge pointer to indicate 150 bar at 20 °C.

NOTE:

Argon pressure level inside the extinguishing agent container may be influenced by ambient temperature, the pressure gauge may indicate higher or lower than 150 bar standard level, please refer to Schedule 1 for the relevant data. If you have any questions, please contact Protecfire.

Schedule 1: Under ideal conditions, here is a table showing the relationship between temperature and pressure of argon at constant volume.



Annual maintenance

Implementation: Protecfire or personnel trained and certified by Protecfire.

The program for annual maintenance can be found on the following pages. The annual maintenance result and record must be conveyed to Protecfire.

Note the following instructions for implementation:

- Pressure test - Control line and Extinguishing line.

NOTE:

Check components are all in place and well secured on pipe clamps even if the pressure test is passed.

After 5 years: Components to be replaced

Implementation: Protecfire or personnel trained and certified by Protecfire.

The following components (if present) must be replaced during this maintenance cycle:

- Extinguishing agent
- SPY fire detection elements
- Pneumatic manual actuator
- E-SPY electric manual actuator
- Pressure switch
- Directional control valve
- Booster
- Hoses
- Pipe clamps
- Band clamps for extinguishing container

After 10 years: Components to be replaced

Implementation: Protecfire or personnel trained and certified by Protecfire.

The following components (if present) must be replaced during this maintenance cycle:

- Extinguishing agent container
- Extinguishing agent
- SPY fire detection elements
- Pneumatic manual actuator
- E-SPY electric manual actuator
- Pressure switch
- Nozzles
- Directional control valve
- Booster
- Hoses
- Pipe clamps
- Band clamps for extinguishing container

Order the components from Protecfire about 6 weeks in advance before the end of 5-year and 10-year period. The part numbers of the components can be found in the chapter under *Installed components*. You will receive a new technical manual with the new extinguishing agent container. After completing installation, the new and the adopted components must be entered in the new manual in the section *Installed components*.

7.2 Annual maintenance tasks

1. First annual maintenance

Extinguishing agent container

Outwardly undamaged?	Yes ☐	No ☐
Screw connections firmly tightened?	Yes ☐	No ☐
Extinguishing agent container firmly attached?	Yes ☐	No ☐
Band clamp firmly attached?	Yes ☐	No ☐

Pressure gauge

Outwardly undamaged?	Yes ☐	No ☐
The oil level in the pressure gauge is at 3/4 of the dial's diameter?	Yes ☐	No ☐
Screw connection firmly tightened?	Yes ☐	No ☐
Pressure level correct?	Yes ☐	No ☐

Pressure switch

Outwardly undamaged?	Yes ☐	No ☐
Screw connection firmly tightened?	Yes ☐	No ☐
The pressure switch in reset state?	Yes ☐	No ☐

Fire detection elements

Outwardly undamaged?	Yes ☐	No ☐
Heat-sensitive pressure-releasing element visible and undamaged?	Yes ☐	No ☐
Free of contamination?	Yes ☐	No ☐
Screw connections firmly tightened?	Yes ☐	No ☐
Pipe clamp firmly attached?	Yes ☐	No ☐

Nozzles

Outwardly undamaged?	Yes ☐	No ☐
Has the red protective cap been replaced?	Yes ☐	No ☐
Nozzles on protected object aligned according to specifications?	Yes ☐	No ☐
Screw connections firmly tightened?	Yes ☐	No ☐

Pneumatic manual actuator

Outwardly undamaged?	Yes ☐	No ☐
Retaining rings / seals undamaged?	Yes ☐	No ☐
Information signs present and legible?	Yes ☐	No ☐
Screw connections firmly tightened?	Yes ☐	No ☐
Pipe clamp firmly attached?	Yes ☐	No ☐

Electric manual actuator

Outwardly undamaged?	Yes ☐	No ☐
Green indicator mark intact and clearly visible?	Yes ☐	No ☐
Free of contamination?	Yes ☐	No ☐
Screw connections firmly tightened?	Yes ☐	No ☐
Pipe clamp firmly attached?	Yes ☐	No ☐

Booster

Outwardly undamaged?	Yes ☐	No ☐
Green indicator mark intact and clearly visible?	Yes ☐	No ☐
Free of contamination?	Yes ☐	No ☐
Screw connections firmly tightened?	Yes ☐	No ☐
Pipe clamp firmly attached?	Yes ☐	No ☐

Directional control valve

Outwardly undamaged?	Yes ☐	No ☐
Screw connection firmly tightened?	Yes ☐	No ☐
Green indicator mark intact?	Yes ☐	No ☐

Extinguishing line

Outwardly undamaged and not deformed?	Yes ☐	No ☐
Blockage in the pipeline?	Yes ☐	No ☐
Pressure test passed?	Yes ☐	No ☐
Screw connections firmly tightened?	Yes ☐	No ☐
Pipe clamp firmly attached?	Yes ☐	No ☐

Control line

Outwardly undamaged and not deformed?	Yes ☐	No ☐
Blockage in the pipeline?	Yes ☐	No ☐
Pressure test passed?	Yes ☐	No ☐
Screw connections firmly tightened?	Yes ☐	No ☐
Pipe clamp firmly attached?	Yes ☐	No ☐

Summary

Fire suppression system free of defects?	Yes ☐	No ☐
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Contact Protecfire if the fire suppression system is not free of defects. If possible, take photos of the damaged components. Describe the defects found:

Customer's signature:	Date:
Maintainer's signature:	Date:

If components need to be replaced, note the replacement in the *installed components* form.

2. Second annual maintenance

Extinguishing agent container

Outwardly undamaged?	Yes ☐	No ☐
Screw connections firmly tightened?	Yes ☐	No ☐
Extinguishing agent container firmly attached?	Yes ☐	No ☐
Band clamp firmly attached?	Yes ☐	No ☐

Pressure gauge

Outwardly undamaged?	Yes ☐	No ☐
The oil level in the pressure gauge is at 3/4 of the dial's diameter?	Yes ☐	No ☐
Screw connection firmly tightened?	Yes ☐	No ☐
Pressure level correct?	Yes ☐	No ☐

Pressure switch

Outwardly undamaged?	Yes ☐	No ☐
Screw connection firmly tightened?	Yes ☐	No ☐
The pressure switch in reset state?	Yes ☐	No ☐

Fire detection elements

Outwardly undamaged?	Yes ☐	No ☐
Heat-sensitive pressure-releasing element visible and undamaged?	Yes ☐	No ☐
Free of contamination?	Yes ☐	No ☐
Screw connections firmly tightened?	Yes ☐	No ☐
Pipe clamp firmly attached?	Yes ☐	No ☐

Nozzles

Outwardly undamaged?	Yes ☐	No ☐
Has the red protective cap been replaced?	Yes ☐	No ☐
Nozzles on protected object aligned according to specifications?	Yes ☐	No ☐
Screw connections firmly tightened?	Yes ☐	No ☐

Pneumatic manual actuator

Outwardly undamaged?	Yes ☐	No ☐
Retaining rings / seals undamaged?	Yes ☐	No ☐
Information signs present and legible?	Yes ☐	No ☐
Screw connections firmly tightened?	Yes ☐	No ☐
Pipe clamp firmly attached?	Yes ☐	No ☐

Electric manual actuator

Outwardly undamaged?	Yes ☐	No ☐
Green indicator mark intact and clearly visible?	Yes ☐	No ☐
Free of contamination?	Yes ☐	No ☐
Screw connections firmly tightened?	Yes ☐	No ☐
Pipe clamp firmly attached?	Yes ☐	No ☐

Booster

Outwardly undamaged?	Yes ☐	No ☐
Green indicator mark intact and clearly visible?	Yes ☐	No ☐
Free of contamination?	Yes ☐	No ☐
Screw connections firmly tightened?	Yes ☐	No ☐
Pipe clamp firmly attached?	Yes ☐	No ☐

Directional control valve

Outwardly undamaged?	Yes ☐	No ☐
Screw connection firmly tightened?	Yes ☐	No ☐
Green indicator mark intact?	Yes ☐	No ☐

Extinguishing line

Outwardly undamaged and not deformed?	Yes ☐	No ☐
Blockage in the pipeline?	Yes ☐	No ☐
Pressure test passed?	Yes ☐	No ☐
Screw connections firmly tightened?	Yes ☐	No ☐
Pipe clamp firmly attached?	Yes ☐	No ☐

Control line

Outwardly undamaged and not deformed?	Yes ☐	No ☐
Blockage in the pipeline?	Yes ☐	No ☐
Pressure test passed?	Yes ☐	No ☐
Screw connections firmly tightened?	Yes ☐	No ☐
Pipe clamp firmly attached?	Yes ☐	No ☐

Summary

Fire suppression system free of defects?	Yes ☐	No ☐
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Contact Protecfire if the fire suppression system is not free of defects. If possible, take photos of the damaged components. Describe the defects found:

Customer's signature:	Date:
Maintainer's signature:	Date:

If components need to be replaced, note the replacement in the *installed components* form.

3. Third annual maintenance

Extinguishing agent container

Outwardly undamaged?	Yes ☐	No ☐
Screw connections firmly tightened?	Yes ☐	No ☐
Extinguishing agent container firmly attached?	Yes ☐	No ☐
Band clamp firmly attached?	Yes ☐	No ☐

Pressure gauge

Outwardly undamaged?	Yes ☐	No ☐
The oil level in the pressure gauge is at 3/4 of the dial's diameter?	Yes ☐	No ☐
Screw connection firmly tightened?	Yes ☐	No ☐
Pressure level correct?	Yes ☐	No ☐

Pressure switch

Outwardly undamaged?	Yes ☐	No ☐
Screw connection firmly tightened?	Yes ☐	No ☐
The pressure switch in reset state?	Yes ☐	No ☐

Fire detection elements

Outwardly undamaged?	Yes ☐	No ☐
Heat-sensitive pressure-releasing element visible and undamaged?	Yes ☐	No ☐
Free of contamination?	Yes ☐	No ☐
Screw connections firmly tightened?	Yes ☐	No ☐
Pipe clamp firmly attached?	Yes ☐	No ☐

Nozzles

Outwardly undamaged?	Yes ☐	No ☐
Has the red protective cap been replaced?	Yes ☐	No ☐
Nozzles on protected object aligned according to specifications?	Yes ☐	No ☐
Screw connections firmly tightened?	Yes ☐	No ☐

Pneumatic manual actuator

Outwardly undamaged?	Yes ☐	No ☐
Retaining rings / seals undamaged?	Yes ☐	No ☐
Information signs present and legible?	Yes ☐	No ☐
Screw connections firmly tightened?	Yes ☐	No ☐
Pipe clamp firmly attached?	Yes ☐	No ☐

Electric manual actuator

Outwardly undamaged?	Yes ☐	No ☐
Green indicator mark intact and clearly visible?	Yes ☐	No ☐
Free of contamination?	Yes ☐	No ☐
Screw connections firmly tightened?	Yes ☐	No ☐
Pipe clamp firmly attached?	Yes ☐	No ☐

Booster

Outwardly undamaged?	Yes ☐	No ☐
Green indicator mark intact and clearly visible?	Yes ☐	No ☐
Free of contamination?	Yes ☐	No ☐
Screw connections firmly tightened?	Yes ☐	No ☐
Pipe clamp firmly attached?	Yes ☐	No ☐

Directional control valve

Outwardly undamaged?	Yes ☐	No ☐
Screw connection firmly tightened?	Yes ☐	No ☐
Green indicator mark intact?	Yes ☐	No ☐

Extinguishing line

Outwardly undamaged and not deformed?	Yes ☐	No ☐
Blockage in the pipeline?	Yes ☐	No ☐
Pressure test passed?	Yes ☐	No ☐
Screw connections firmly tightened?	Yes ☐	No ☐
Pipe clamp firmly attached?	Yes ☐	No ☐

Control line

Outwardly undamaged and not deformed?	Yes ☐	No ☐
Blockage in the pipeline?	Yes ☐	No ☐
Pressure test passed?	Yes ☐	No ☐
Screw connections firmly tightened?	Yes ☐	No ☐
Pipe clamp firmly attached?	Yes ☐	No ☐

Summary

Fire suppression system free of defects?	Yes ☐	No ☐
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Contact Protecfire if the fire suppression system is not free of defects. If possible, take photos of the damaged components. Describe the defects found:

Customer's signature:	Date:
Maintainer's signature:	Date:

If components need to be replaced, note the replacement in the *installed components* form.

4. Fourth annual maintenance

Extinguishing agent container

Outwardly undamaged?	Yes ☐	No ☐
Screw connections firmly tightened?	Yes ☐	No ☐
Extinguishing agent container firmly attached?	Yes ☐	No ☐
Band clamp firmly attached?	Yes ☐	No ☐

Pressure gauge

Outwardly undamaged?	Yes ☐	No ☐
The oil level in the pressure gauge is at 3/4 of the dial's diameter?	Yes ☐	No ☐
Screw connection firmly tightened?	Yes ☐	No ☐
Pressure level correct?	Yes ☐	No ☐

Pressure switch

Outwardly undamaged?	Yes ☐	No ☐
Screw connection firmly tightened?	Yes ☐	No ☐
The pressure switch in reset state?	Yes ☐	No ☐

Fire detection elements

Outwardly undamaged?	Yes ☐	No ☐
Heat-sensitive pressure-releasing element visible and undamaged?	Yes ☐	No ☐
Free of contamination?	Yes ☐	No ☐
Screw connections firmly tightened?	Yes ☐	No ☐
Pipe clamp firmly attached?	Yes ☐	No ☐

Nozzles

Outwardly undamaged?	Yes ☐	No ☐
Has the red protective cap been replaced?	Yes ☐	No ☐
Nozzles on protected object aligned according to specifications?	Yes ☐	No ☐
Screw connections firmly tightened?	Yes ☐	No ☐

Pneumatic manual actuator

Outwardly undamaged?	Yes ☐	No ☐
Retaining rings / seals undamaged?	Yes ☐	No ☐
Information signs present and legible?	Yes ☐	No ☐
Screw connections firmly tightened?	Yes ☐	No ☐
Pipe clamp firmly attached?	Yes ☐	No ☐

Electric manual actuator

Outwardly undamaged?	Yes ☐	No ☐
Green indicator mark intact and clearly visible?	Yes ☐	No ☐
Free of contamination?	Yes ☐	No ☐
Screw connections firmly tightened?	Yes ☐	No ☐
Pipe clamp firmly attached?	Yes ☐	No ☐

Booster

Outwardly undamaged?	Yes ☐	No ☐
Green indicator mark intact and clearly visible?	Yes ☐	No ☐
Free of contamination?	Yes ☐	No ☐
Screw connections firmly tightened?	Yes ☐	No ☐
Pipe clamp firmly attached?	Yes ☐	No ☐

Directional control valve

Outwardly undamaged?	Yes ☐	No ☐
Screw connection firmly tightened?	Yes ☐	No ☐
Green indicator mark intact?	Yes ☐	No ☐

Extinguishing line

Outwardly undamaged and not deformed?	Yes ☐	No ☐
Blockage in the pipeline?	Yes ☐	No ☐
Pressure test passed?	Yes ☐	No ☐
Screw connections firmly tightened?	Yes ☐	No ☐
Pipe clamp firmly attached?	Yes ☐	No ☐

Control line

Outwardly undamaged and not deformed?	Yes ☐	No ☐
Blockage in the pipeline?	Yes ☐	No ☐
Pressure test passed?	Yes ☐	No ☐
Screw connections firmly tightened?	Yes ☐	No ☐
Pipe clamp firmly attached?	Yes ☐	No ☐

Summary

Fire suppression system free of defects?	Yes ☐	No ☐
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Contact Protecfire if the fire suppression system is not free of defects. If possible, take photos of the damaged components. Describe the defects found:

Customer's signature:	Date:
Maintainer's signature:	Date:

If components need to be replaced, note the replacement in the *installed components* form.

5. Fifth maintenance cycle: replacement components

After a period of five years, the listed components must be replaced if there are any. Other components must be replaced if an inspection reveals defects. Protecfire recommends ordering all replacement components from Protecfire approximately 6 weeks before the end of the five-year period. The part numbers of the components can be found in the section “Installed Components”. After the components have been replaced, commissioning must be performed again. This process must be documented in the Commissioning checklist (Maintenance).

Replaced components	Part no.	Quantity	Serial number
Extinguishing agent			
Pneumatic manual actuator	84012000		
E-SPY electric manual actuator	84107900		
Pressure switch	86006026		
Directional control valve	80108050		
Booster 380	84210000		
SPY fire detection element _____ °C			
Hoses			
Pipe clamps			
Band clamps			

Other replaced components (please enter):

Customer's signature: _____ Date: _____

Maintainer's signature: _____ Date: _____

Dispose of old parts in accordance with national regulations. If you have any questions about the disposal of old parts, please contact Protecfire.

5.1 Commissioning checklist (Maintenance)

Once all components have been assembled and the Pressure test has passed, the fire suppression system can be operated normally. The checklist below is used to make a final check that the installation is correct. The fire suppression system can be handed over if all questions are answered with "Yes". Otherwise, the defect must be remedied before handover.

Extinguishing agent container

Outwardly undamaged?	Yes ☐	No ☐
Screw connections firmly tightened?	Yes ☐	No ☐
Extinguishing agent container firmly attached?	Yes ☐	No ☐
Band clamp firmly attached?	Yes ☐	No ☐

Pressure gauge

Outwardly undamaged?	Yes ☐	No ☐
The oil level in the pressure gauge is at 3/4 of the dial's diameter?	Yes ☐	No ☐
Screw connections firmly tightened?	Yes ☐	No ☐
Pressure level correct?	Yes ☐	No ☐

Pressure switch

Outwardly undamaged?	Yes ☐	No ☐
Screw connection firmly tightened?	Yes ☐	No ☐
The pressure switch in reset state?	Yes ☐	No ☐

Fire detection elements

Outwardly undamaged?	Yes ☐	No ☐
Installation in protected area?	Yes ☐	No ☐
Activation temperatures according to design?	Yes ☐	No ☐
Free of contamination?	Yes ☐	No ☐
Screw connections firmly tightened?	Yes ☐	No ☐
Pipe clamp firmly attached?	Yes ☐	No ☐

Nozzles

Outwardly undamaged?	Yes ☐	No ☐
Has the red protective cap been replaced?	Yes ☐	No ☐
Nozzles on protected object aligned according to specifications?	Yes ☐	No ☐
Screw connections firmly tightened?	Yes ☐	No ☐

Pneumatic manual actuator

Outwardly undamaged?	Yes ☐	No ☐
Easily accessible?	Yes ☐	No ☐
Mounted outside of fire hazard areas?	Yes ☐	No ☐
Retaining rings / seals undamaged?	Yes ☐	No ☐
Retaining rings free of contamination?	Yes ☐	No ☐
Information signs present and legible?	Yes ☐	No ☐
Screw connections firmly tightened?	Yes ☐	No ☐
Pipe clamp firmly attached?	Yes ☐	No ☐

Electric manual actuator

Outwardly undamaged?	Yes ☐	No ☐
Installation in specified area?	Yes ☐	No ☐
Green indicator mark intact and clearly visible?	Yes ☐	No ☐
Free of contamination?	Yes ☐	No ☐
Screw connections firmly tightened?	Yes ☐	No ☐
Pipe clamp firmly attached?	Yes ☐	No ☐

Booster

Outwardly undamaged?	Yes ☐	No ☐
Installation in specified area?	Yes ☐	No ☐
Green indicator mark intact and clearly visible?	Yes ☐	No ☐
Free of contamination?	Yes ☐	No ☐
Screw connections firmly tightened?	Yes ☐	No ☐
Pipe clamp firmly attached?	Yes ☐	No ☐

Directional control valve

Outwardly undamaged?	Yes ☐	No ☐
Screw connection firmly tightened?	Yes ☐	No ☐
Green indicator mark intact?	Yes ☐	No ☐

Extinguishing line

Outwardly undamaged and not deformed?	Yes ☐	No ☐
Blockage in the pipeline?	Yes ☐	No ☐
Pressure test passed?	Yes ☐	No ☐
Distance between pipe clamps a maximum of 0.6 m?	Yes ☐	No ☐
Screw connections firmly tightened?	Yes ☐	No ☐
Pipe clamp firmly attached?	Yes ☐	No ☐

Control line

Outwardly undamaged and not deformed?	Yes ☐	No ☐
Blockage in the pipeline?	Yes ☐	No ☐
Pressure test passed?	Yes ☐	No ☐
Distance between pipe clamps a maximum of 0.6 m?	Yes ☐	No ☐
Screw connections firmly tightened?	Yes ☐	No ☐
Pipe clamp firmly attached?	Yes ☐	No ☐

Summary

Fire suppression system free of defects?	Yes ☐	No ☐
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Comment

Customer's signature:	Date:
Maintainer's signature:	Date:

6. Sixth annual maintenance

Extinguishing agent container

Outwardly undamaged?	Yes ☐	No ☐
Screw connections firmly tightened?	Yes ☐	No ☐
Extinguishing agent container firmly attached?	Yes ☐	No ☐
Band clamp firmly attached?	Yes ☐	No ☐

Pressure gauge

Outwardly undamaged?	Yes ☐	No ☐
The oil level in the pressure gauge is at 3/4 of the dial's diameter?	Yes ☐	No ☐
Screw connection firmly tightened?	Yes ☐	No ☐
Pressure level correct?	Yes ☐	No ☐

Pressure switch

Outwardly undamaged?	Yes ☐	No ☐
Screw connection firmly tightened?	Yes ☐	No ☐
The pressure switch in reset state?	Yes ☐	No ☐

Fire detection elements

Outwardly undamaged?	Yes ☐	No ☐
Heat-sensitive pressure-releasing element visible and undamaged?	Yes ☐	No ☐
Free of contamination?	Yes ☐	No ☐
Screw connections firmly tightened?	Yes ☐	No ☐
Pipe clamp firmly attached?	Yes ☐	No ☐

Nozzles

Outwardly undamaged?	Yes ☐	No ☐
Has the red protective cap been replaced?	Yes ☐	No ☐
Nozzles on protected object aligned according to specifications?	Yes ☐	No ☐
Screw connections firmly tightened?	Yes ☐	No ☐

Pneumatic manual actuator

Outwardly undamaged?	Yes ☐	No ☐
Retaining rings / seals undamaged?	Yes ☐	No ☐
Information signs present and legible?	Yes ☐	No ☐
Screw connections firmly tightened?	Yes ☐	No ☐
Pipe clamp firmly attached?	Yes ☐	No ☐

Electric manual actuator

Outwardly undamaged?	Yes ☐	No ☐
Green indicator mark intact and clearly visible?	Yes ☐	No ☐
Free of contamination?	Yes ☐	No ☐
Screw connections firmly tightened?	Yes ☐	No ☐
Pipe clamp firmly attached?	Yes ☐	No ☐

Booster

Outwardly undamaged?	Yes ☐	No ☐
Green indicator mark intact and clearly visible?	Yes ☐	No ☐
Free of contamination?	Yes ☐	No ☐
Screw connections firmly tightened?	Yes ☐	No ☐
Pipe clamp firmly attached?	Yes ☐	No ☐

Directional control valve

Outwardly undamaged?	Yes ☐	No ☐
Screw connection firmly tightened?	Yes ☐	No ☐
Green indicator mark intact?	Yes ☐	No ☐

Extinguishing line

Outwardly undamaged and not deformed?	Yes ☐	No ☐
Blockage in the pipeline?	Yes ☐	No ☐
Pressure test passed?	Yes ☐	No ☐
Screw connections firmly tightened?	Yes ☐	No ☐
Pipe clamp firmly attached?	Yes ☐	No ☐

Control line

Outwardly undamaged and not deformed?	Yes ☐	No ☐
Blockage in the pipeline?	Yes ☐	No ☐
Pressure test passed?	Yes ☐	No ☐
Screw connections firmly tightened?	Yes ☐	No ☐
Pipe clamp firmly attached?	Yes ☐	No ☐

Summary

Fire suppression system free of defects?	Yes ☐	No ☐
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Contact Protecfire if the fire suppression system is not free of defects. If possible, take photos of the damaged components. Describe the defects found:

Customer's signature:	Date:
Maintainer's signature:	Date:

If components need to be replaced, note the replacement in the *installed components* form.

7. Seventh annual maintenance

Extinguishing agent container

Outwardly undamaged?	Yes ☐	No ☐
Screw connections firmly tightened?	Yes ☐	No ☐
Extinguishing agent container firmly attached?	Yes ☐	No ☐
Band clamp firmly attached?	Yes ☐	No ☐

Pressure gauge

Outwardly undamaged?	Yes ☐	No ☐
The oil level in the pressure gauge is at 3/4 of the dial's diameter?	Yes ☐	No ☐
Screw connection firmly tightened?	Yes ☐	No ☐
Pressure level correct?	Yes ☐	No ☐

Pressure switch

Outwardly undamaged?	Yes ☐	No ☐
Screw connection firmly tightened?	Yes ☐	No ☐
The pressure switch in reset state?	Yes ☐	No ☐

Fire detection elements

Outwardly undamaged?	Yes ☐	No ☐
Heat-sensitive pressure-releasing element visible and undamaged?	Yes ☐	No ☐
Free of contamination?	Yes ☐	No ☐
Screw connections firmly tightened?	Yes ☐	No ☐
Pipe clamp firmly attached?	Yes ☐	No ☐

Nozzles

Outwardly undamaged?	Yes ☐	No ☐
Has the red protective cap been replaced?	Yes ☐	No ☐
Nozzles on protected object aligned according to specifications?	Yes ☐	No ☐
Screw connections firmly tightened?	Yes ☐	No ☐

Pneumatic manual actuator

Outwardly undamaged?	Yes ☐	No ☐
Retaining rings / seals undamaged?	Yes ☐	No ☐
Information signs present and legible?	Yes ☐	No ☐
Screw connections firmly tightened?	Yes ☐	No ☐
Pipe clamp firmly attached?	Yes ☐	No ☐

Electric manual actuator

Outwardly undamaged?	Yes ☐	No ☐
Green indicator mark intact and clearly visible?	Yes ☐	No ☐
Free of contamination?	Yes ☐	No ☐
Screw connections firmly tightened?	Yes ☐	No ☐
Pipe clamp firmly attached?	Yes ☐	No ☐

Booster

Outwardly undamaged?	Yes ☐	No ☐
Green indicator mark intact and clearly visible?	Yes ☐	No ☐
Free of contamination?	Yes ☐	No ☐
Screw connections firmly tightened?	Yes ☐	No ☐
Pipe clamp firmly attached?	Yes ☐	No ☐

Directional control valve

Outwardly undamaged?	Yes ☐	No ☐
Screw connection firmly tightened?	Yes ☐	No ☐
Green indicator mark intact?	Yes ☐	No ☐

Extinguishing line

Outwardly undamaged and not deformed?	Yes ☐	No ☐
Blockage in the pipeline?	Yes ☐	No ☐
Pressure test passed?	Yes ☐	No ☐
Screw connections firmly tightened?	Yes ☐	No ☐
Pipe clamp firmly attached?	Yes ☐	No ☐

Control line

Outwardly undamaged and not deformed?	Yes ☐	No ☐
Blockage in the pipeline?	Yes ☐	No ☐
Pressure test passed?	Yes ☐	No ☐
Screw connections firmly tightened?	Yes ☐	No ☐
Pipe clamp firmly attached?	Yes ☐	No ☐

Summary

Fire suppression system free of defects?	Yes ☐	No ☐
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Contact Protecfire if the fire suppression system is not free of defects. If possible, take photos of the damaged components. Describe the defects found:

Customer's signature:	Date:
Maintainer's signature:	Date:

If components need to be replaced, note the replacement in the *installed components* form.

8. Eighth annual maintenance

Extinguishing agent container

Outwardly undamaged?	Yes ☐	No ☐
Screw connections firmly tightened?	Yes ☐	No ☐
Extinguishing agent container firmly attached?	Yes ☐	No ☐
Band clamp firmly attached?	Yes ☐	No ☐

Pressure gauge

Outwardly undamaged?	Yes ☐	No ☐
The oil level in the pressure gauge is at 3/4 of the dial's diameter?	Yes ☐	No ☐
Screw connection firmly tightened?	Yes ☐	No ☐
Pressure level correct?	Yes ☐	No ☐

Pressure switch

Outwardly undamaged?	Yes ☐	No ☐
Screw connection firmly tightened?	Yes ☐	No ☐
The pressure switch in reset state?	Yes ☐	No ☐

Fire detection elements

Outwardly undamaged?	Yes ☐	No ☐
Heat-sensitive pressure-releasing element visible and undamaged?	Yes ☐	No ☐
Free of contamination?	Yes ☐	No ☐
Screw connections firmly tightened?	Yes ☐	No ☐
Pipe clamp firmly attached?	Yes ☐	No ☐

Nozzles

Outwardly undamaged?	Yes ☐	No ☐
Has the red protective cap been replaced?	Yes ☐	No ☐
Nozzles on protected object aligned according to specifications?	Yes ☐	No ☐
Screw connections firmly tightened?	Yes ☐	No ☐

Pneumatic manual actuator

Outwardly undamaged?	Yes ☐	No ☐
Retaining rings / seals undamaged?	Yes ☐	No ☐
Information signs present and legible?	Yes ☐	No ☐
Screw connections firmly tightened?	Yes ☐	No ☐
Pipe clamp firmly attached?	Yes ☐	No ☐

Electric manual actuator

Outwardly undamaged?	Yes ☐	No ☐
Green indicator mark intact and clearly visible?	Yes ☐	No ☐
Free of contamination?	Yes ☐	No ☐
Screw connections firmly tightened?	Yes ☐	No ☐
Pipe clamp firmly attached?	Yes ☐	No ☐

Booster

Outwardly undamaged?	Yes ☐	No ☐
Green indicator mark intact and clearly visible?	Yes ☐	No ☐
Free of contamination?	Yes ☐	No ☐
Screw connections firmly tightened?	Yes ☐	No ☐
Pipe clamp firmly attached?	Yes ☐	No ☐

Directional control valve

Outwardly undamaged?	Yes ☐	No ☐
Screw connection firmly tightened?	Yes ☐	No ☐
Green indicator mark intact?	Yes ☐	No ☐

Extinguishing line

Outwardly undamaged and not deformed?	Yes ☐	No ☐
Blockage in the pipeline?	Yes ☐	No ☐
Pressure test passed?	Yes ☐	No ☐
Screw connections firmly tightened?	Yes ☐	No ☐
Pipe clamp firmly attached?	Yes ☐	No ☐

Control line

Outwardly undamaged and not deformed?	Yes ☐	No ☐
Blockage in the pipeline?	Yes ☐	No ☐
Pressure test passed?	Yes ☐	No ☐
Screw connections firmly tightened?	Yes ☐	No ☐
Pipe clamp firmly attached?	Yes ☐	No ☐

Summary

Fire suppression system free of defects?	Yes ☐	No ☐
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Contact Protecfire if the fire suppression system is not free of defects. If possible, take photos of the damaged components. Describe the defects found:

Customer's signature:	Date:
Maintainer's signature:	Date:

If components need to be replaced, note the replacement in the *installed components* form.

9. Ninth annual maintenance

Extinguishing agent container

Outwardly undamaged?	Yes ☐	No ☐
Screw connections firmly tightened?	Yes ☐	No ☐
Extinguishing agent container firmly attached?	Yes ☐	No ☐
Band clamp firmly attached?	Yes ☐	No ☐

Pressure gauge

Outwardly undamaged?	Yes ☐	No ☐
The oil level in the pressure gauge is at 3/4 of the dial's diameter?	Yes ☐	No ☐
Screw connection firmly tightened?	Yes ☐	No ☐
Pressure level correct?	Yes ☐	No ☐

Pressure switch

Outwardly undamaged?	Yes ☐	No ☐
Screw connection firmly tightened?	Yes ☐	No ☐
The pressure switch in reset state?	Yes ☐	No ☐

Fire detection elements

Outwardly undamaged?	Yes ☐	No ☐
Heat-sensitive pressure-releasing element visible and undamaged?	Yes ☐	No ☐
Free of contamination?	Yes ☐	No ☐
Screw connections firmly tightened?	Yes ☐	No ☐
Pipe clamp firmly attached?	Yes ☐	No ☐

Nozzles

Outwardly undamaged?	Yes ☐	No ☐
Has the red protective cap been replaced?	Yes ☐	No ☐
Nozzles on protected object aligned according to specifications?	Yes ☐	No ☐
Screw connections firmly tightened?	Yes ☐	No ☐

Pneumatic manual actuator

Outwardly undamaged?	Yes ☐	No ☐
Retaining rings / seals undamaged?	Yes ☐	No ☐
Information signs present and legible?	Yes ☐	No ☐
Screw connections firmly tightened?	Yes ☐	No ☐
Pipe clamp firmly attached?	Yes ☐	No ☐

Electric manual actuator

Outwardly undamaged?	Yes ☐	No ☐
Green indicator mark intact and clearly visible?	Yes ☐	No ☐
Free of contamination?	Yes ☐	No ☐
Screw connections firmly tightened?	Yes ☐	No ☐
Pipe clamp firmly attached?	Yes ☐	No ☐

Booster

Outwardly undamaged?	Yes ☐	No ☐
Green indicator mark intact and clearly visible?	Yes ☐	No ☐
Free of contamination?	Yes ☐	No ☐
Screw connections firmly tightened?	Yes ☐	No ☐
Pipe clamp firmly attached?	Yes ☐	No ☐

Directional control valve

Outwardly undamaged?	Yes ☐	No ☐
Screw connection firmly tightened?	Yes ☐	No ☐
Green indicator mark intact?	Yes ☐	No ☐

Extinguishing line

Outwardly undamaged and not deformed?	Yes ☐	No ☐
Blockage in the pipeline?	Yes ☐	No ☐
Pressure test passed?	Yes ☐	No ☐
Screw connections firmly tightened?	Yes ☐	No ☐
Pipe clamp firmly attached?	Yes ☐	No ☐

Control line

Outwardly undamaged and not deformed?	Yes ☐	No ☐
Blockage in the pipeline?	Yes ☐	No ☐
Pressure test passed?	Yes ☐	No ☐
Screw connections firmly tightened?	Yes ☐	No ☐
Pipe clamp firmly attached?	Yes ☐	No ☐

Summary

Fire suppression system free of defects?	Yes ☐	No ☐
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Contact Protecfire if the fire suppression system is not free of defects. If possible, take photos of the damaged components. Describe the defects found:

Customer's signature:	Date:
Maintainer's signature:	Date:

If components need to be replaced, note the replacement in the *installed components* form.

10. Tenth maintenance cycle: replacement components

After a period of ten years, the listed components must be replaced if there are any. Other components must be replaced if an inspection reveals defects. Protecfire recommends ordering all replacement components from Protecfire approximately 6 weeks before the end of the ten-year period. The part numbers of the components can be found in the section “Installed Components.” The new Extinguishing agent container is equipped with the new technical manual. After the components have been replaced, commissioning must be performed again. This process must be documented in the new technical manual.

Replaced components

Extinguishing agent container	Yes ☐	No ☐
Extinguishing agent	Yes ☐	No ☐
SPY fire detection elements	Yes ☐	No ☐
Pneumatic manual actuator	Yes ☐	No ☐
E-SPY electric manual actuator	Yes ☐	No ☐
Nozzles	Yes ☐	No ☐
Pressure switch	Yes ☐	No ☐
Directional control valves	Yes ☐	No ☐
Booster	Yes ☐	No ☐
Hoses	Yes ☐	No ☐
Pipe clamps	Yes ☐	No ☐
Band clamps	Yes ☐	No ☐

Other replaced components (please enter):

Replaced and accepted components entered in the new manual under *Installed components*? Yes ☐ No ☐

Customer's signature: _____ Date: _____

Maintainer's signature: _____ Date: _____

Dispose of old parts in accordance with national regulations. If you have any questions about the disposal of old parts, please contact Protecfire.

8 Recommissioning after the system has been activated

After the system has been activated, the following tasks must be carried out to ensure that the fire suppression system is ready for operation again as quickly as possible.

Component replacement

Implementation: Protecfire or personnel trained and certified by Protecfire.

The following components, if present, must be replaced:

- Extinguishing agent
- SPY fire detection elements that have been activated
- Pneumatic manual actuator that has been activated
- E-SPY electric manual actuator that has been activated
- Directional control valve that has been activated
- Booster that has been activated
- Nozzles that have been exposed to fire
- Nozzles and pipeline that have been clogged with dirt

Procedure for recommissioning

Implementation: Protecfire or personnel trained and certified by Protecfire.

1. Reset the pressure switch with the button (the alarm signal will be switch off).
2. Disconnect the extinguishing agent container from the extinguishing and control line (hose).
3. Seal extinguishing line and control line with plugs to protect against contamination.
4. Disassemble activated extinguishing agent container and return to the factory for servicing or replace for a new one.
5. Replace activated pneumatic manual actuator (if applicable).
6. Visually inspect the extinguishing line and replace activated SPY fire detection element.
7. Replace activated E-SPY electric manual actuator (if applicable).
8. Replace activated Booster (if applicable).
9. Replace nozzles that have been exposed to fire.
10. Inspect all lines and replace nozzles or pipeline that have become clogged with dirt.
11. Replace directional control valve that has been activated (if applicable).

12. Install the extinguishing agent container.
 13. Remove the plugs from the control line and extinguishing line and connect hoses to extinguishing agent container.
 14. Perform pressure test successfully.
 15. Check and document installation in accordance with the commissioning checklist (Initial installation) in the new technical manual.
 16. Enter replaced and accepted components in the new technical manual in the Installed components form.
- ✓ The fire suppression system is now ready for operation again.

9 Disposal

The components listed in the section Fifth maintenance cycle and Tenth maintenance cycle must be replaced in maintenance. In addition, some components must be replaced if the system is activated. The corresponding specification can be found in the technical data of the components. Dispose of the replaced components in accordance with the regulations of your country. If you have any questions regarding disposal, please contact Protecfire.

Please do not hesitate to contact us should you have any questions.

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